UE24CS352A / MACHINE LEARNING

Corporate Bankruptcy Prediction

Financial ratios, rare outcomes, and a reproducible evaluation

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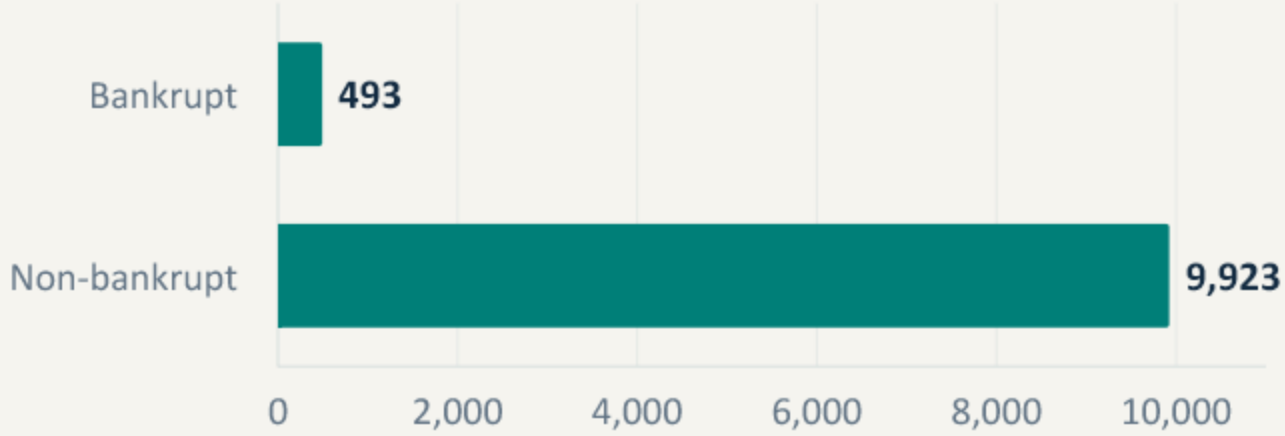
The prediction problem

Bankruptcy within three years

64 financial ratios describe each company statement.
The target is 1 for bankruptcy and 0 otherwise.

4.73%

positive labels after cleaning



An always non-bankrupt model can exceed 95% accuracy while detecting no failures.

Data quality before modeling

Audit item	Verified finding
Raw statements	10,503
Exact duplicate rows removed	87
Clean statements / bankrupt labels	10,416 / 493
Missing predictor cells after cleaning	9,833
Conflicting identical feature vectors	0

Missing values are retained for training-time treatment.
Company IDs are unavailable, so company-level separation cannot be verified.

Experimental design

8,332

development rows

2,084

held-out test rows

Development only	Final evaluation
Three stratified folds	One untouched stratified holdout
Select model by mean average precision	Evaluate the frozen model
Choose threshold by OOF F2	Report detections and errors

Seed 42. Learned preprocessing is fitted separately inside every training fold.

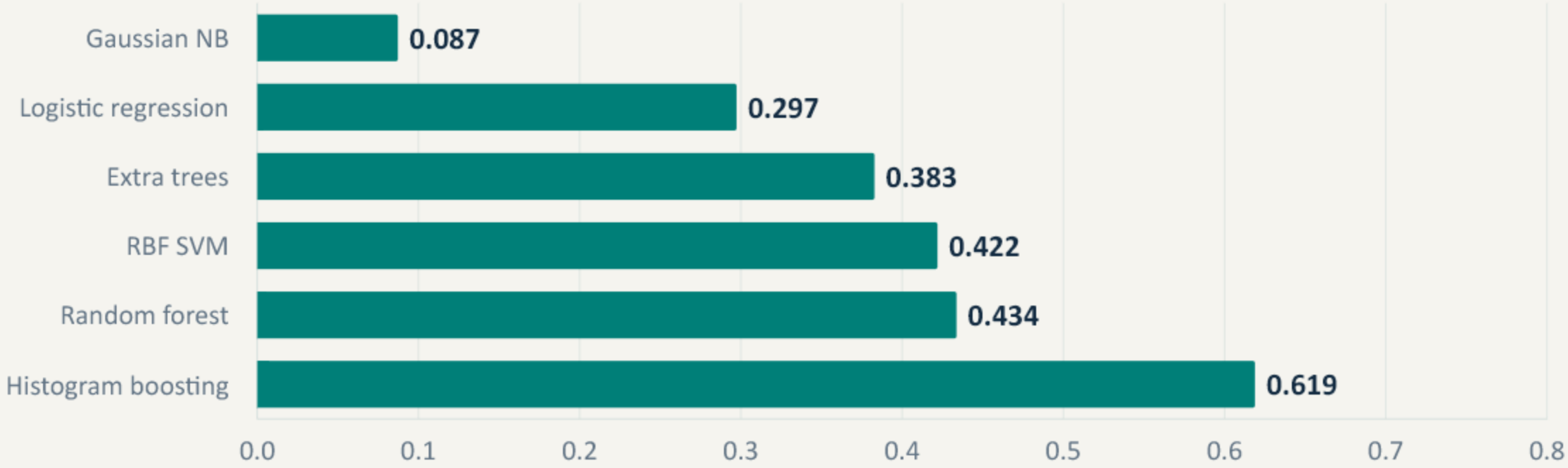
Six classifiers, consistent evaluation

Model family	Preprocessing / imbalance treatment
Logistic regression	Impute, transform, scale; balanced weights
Gaussian naive Bayes	Impute, transform, scale
RBF support vector machine	Impute, transform, scale; balanced weights
Random forest	Impute + missing indicators; balanced weights
Extra trees	Impute + missing indicators; balanced weights
Histogram gradient boosting	Native missing values; balanced weights

Average precision is the primary selection metric because bankruptcy is rare.

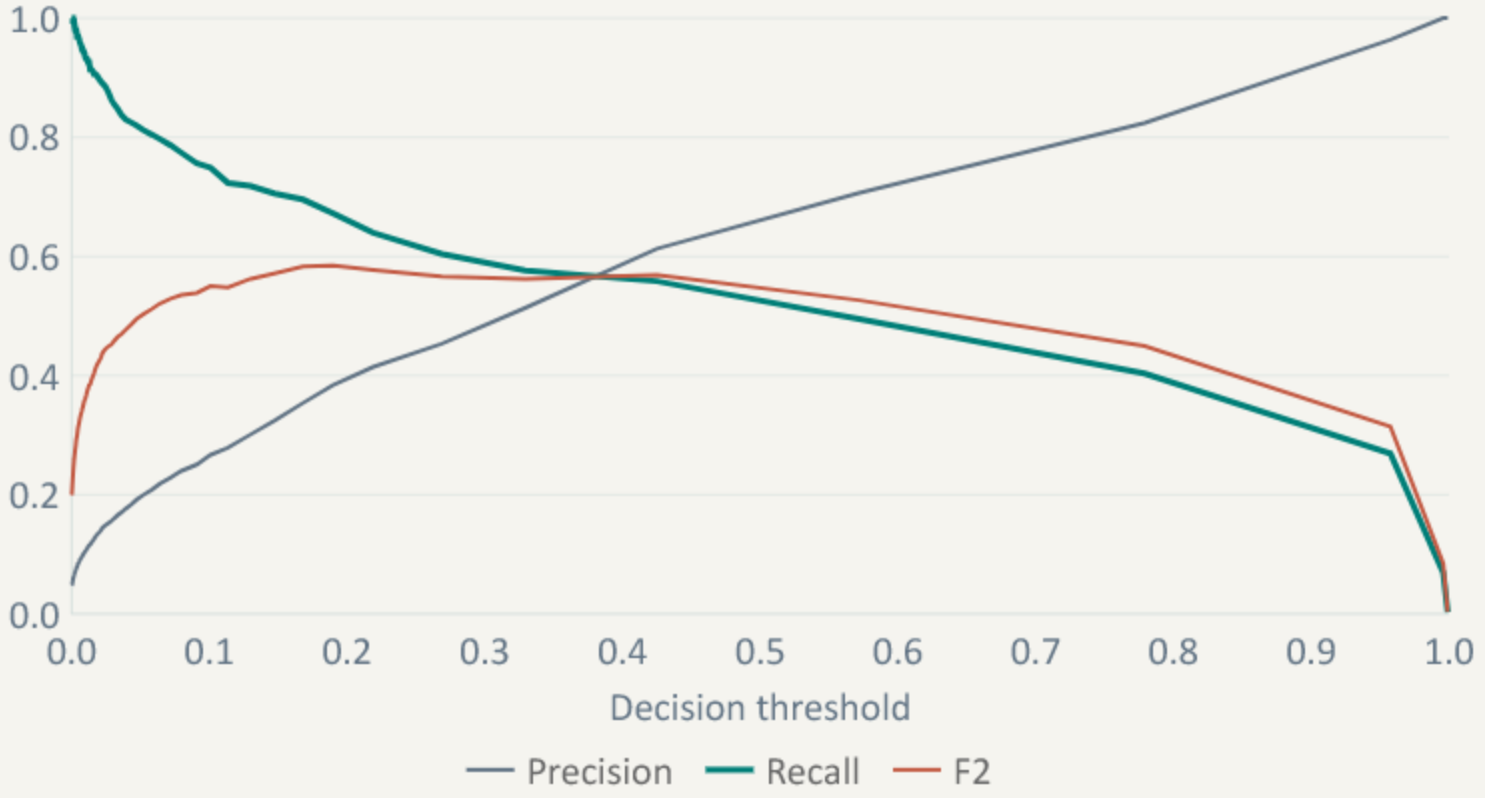
Histogram boosting leads cross-validation

Mean average precision across three development folds



Selected model: mean AP 0.619, fold SD 0.037

The threshold sets the screening trade-off



0.1980

chosen from development out-of-fold predictions

F2 emphasizes recall

The test set does not choose the threshold.

84 of 99 bankruptcies detected

84.8%

bankruptcy recall

34.3%

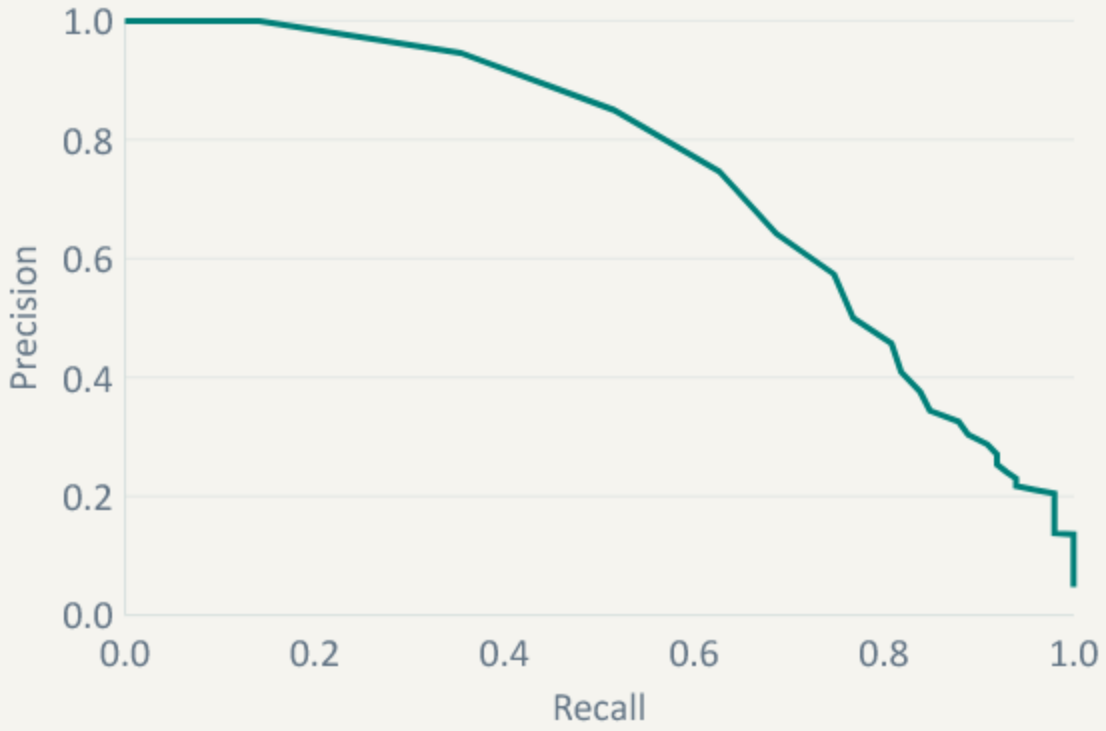
precision

Actual / predicted	Non-bankrupt	Bankrupt
Non-bankrupt	1,824	161
Bankrupt	15	84

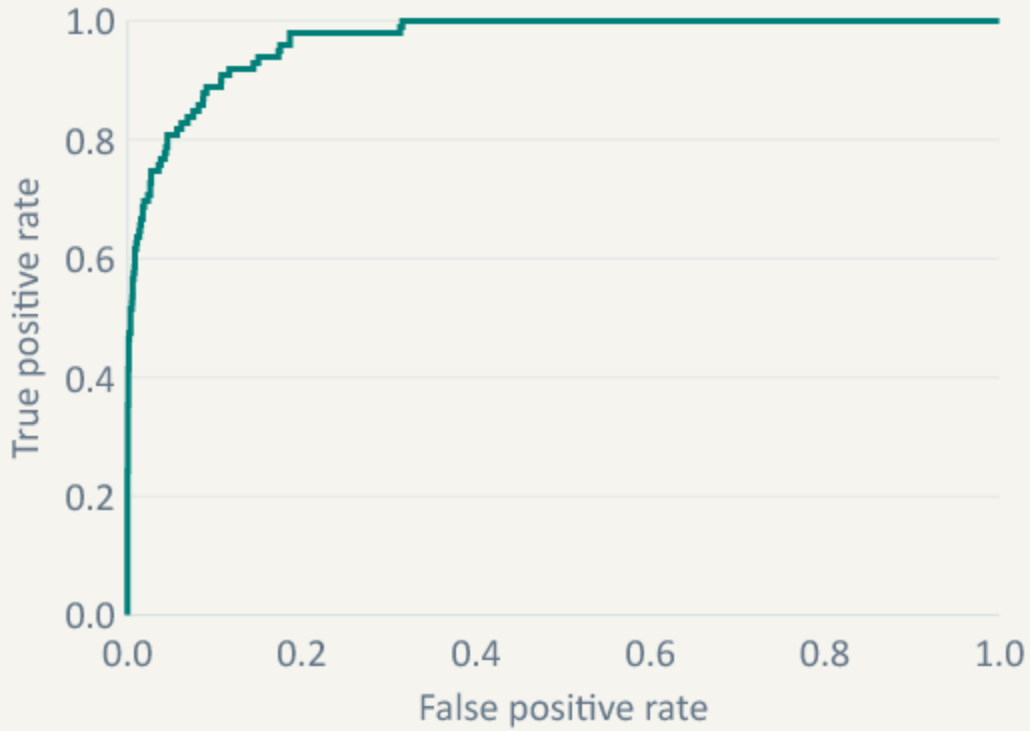
15 missed bankruptcies. 161 false alarms.
Test set: 2,084 statements with 99 positive labels.

Ranking quality on the held-out test set

Average precision 0.750



ROC AUC 0.968



Bootstrap 95% AP interval: 0.683–0.820, conditional on the fixed model and test split.

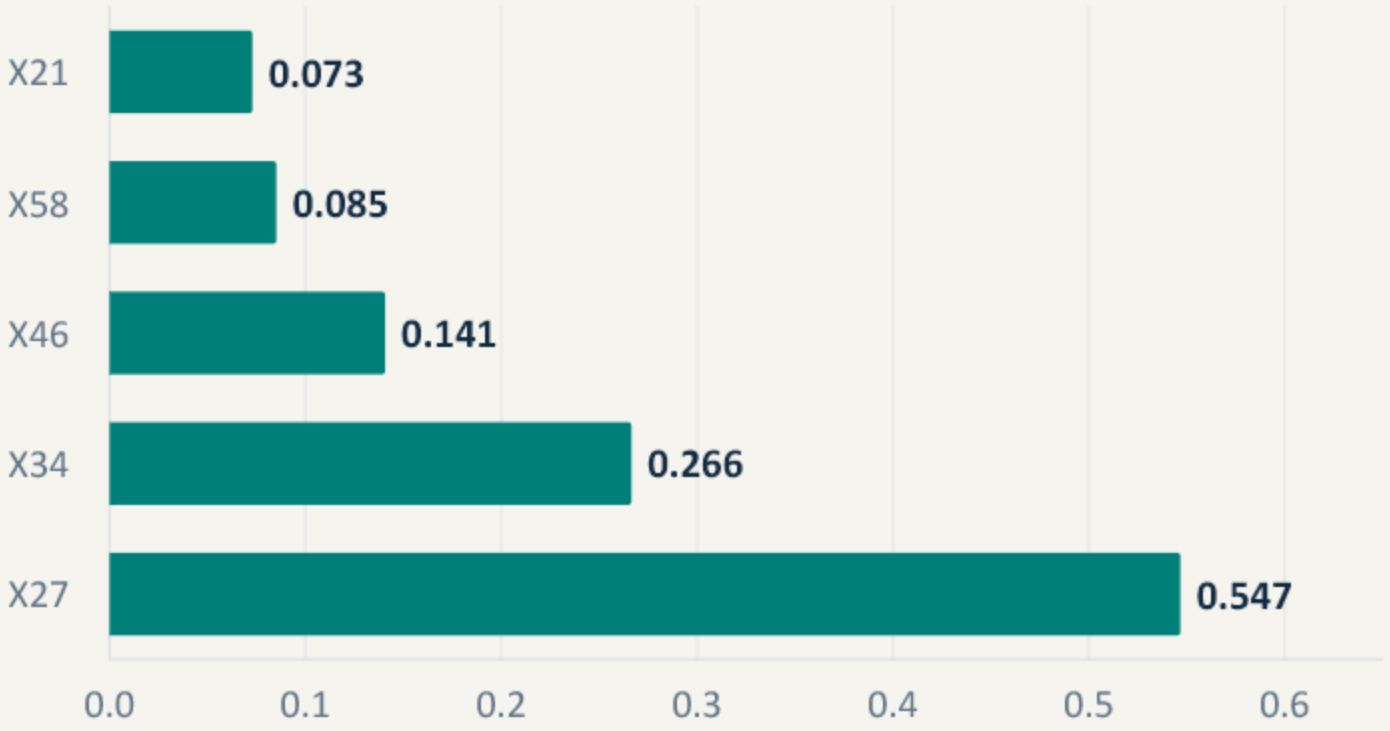
More detections also mean more false alarms

Threshold	Recall	Precision	False alarms	Missed
OOF choice: 0.198	84.8%	34.3%	161	15
Default: 0.500	68.7%	65.4%	36	31

16 additional bankruptcies detected

125 additional false alarms

Which ratios influence the selected model?



X27

Operating profit / financial expenses

X34

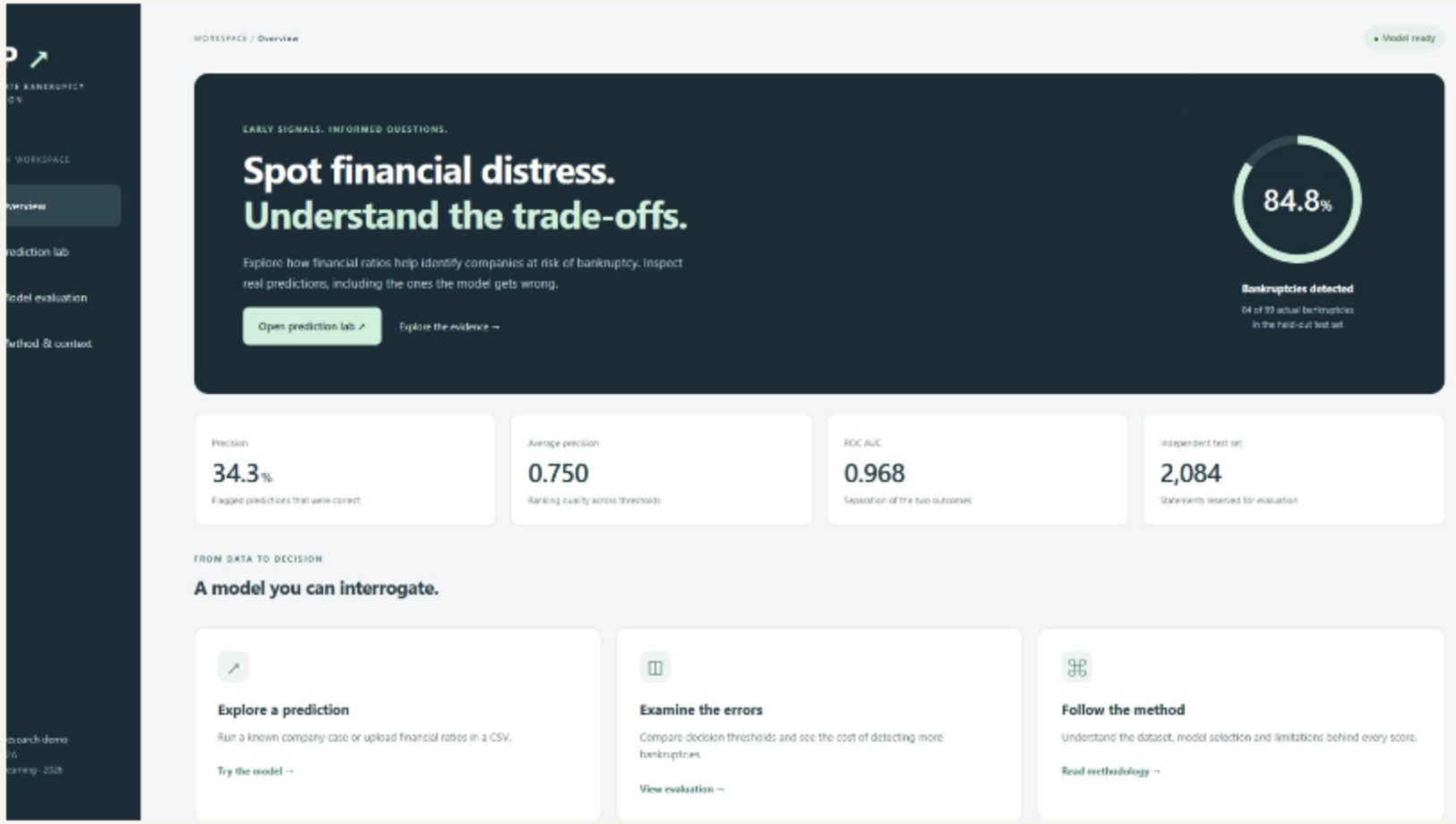
Operating expenses / total liabilities

X46

Quick ratio

Permutation importance measures model reliance. Correlation prevents a causal reading.

Live demonstration



1. Run a known test case
2. Show a missed case or false alarm
3. Upload a CSV and export predictions

python app.py

Conclusion and next steps

A useful signal, with a measurable cost

84 bankruptcies detected
15 missed cases and 161 false alarms

Next validation priorities

Company and date separation
Threshold stability across splits
Calibration on independent data

References and reproducibility

Sources

Tomczak (2016), UCI dataset
doi:10.24432/C5F600

Scikit-learn evaluation and CV documentation

Supplied Patil report and codeatri repository
Background references

Reproduce the experiment

python run.py train
python -m pytest -q

Data hash, split manifest, predictions,
metrics, model and notebook are included in the repository.